Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 20)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 . 1 Explain what is meant by **biomimetics** and describe **three** examples.

[4 marks]

0 1 . 2 Explain why many plants produce chemicals that are useful as human medicines, and give **two** examples.

[3 marks]

0 1 . 3 Explain why the discovery of medicines is a strong argument for conserving whole habitats rather than individual species.

[2 marks]

0 1 . 4 Give **one** example of a wildlife species used in physiological research and the human health issue it helps to understand.

[1 mark]

Do not write outside the box

Do not write outside the box

10

0 2 Table 1 compares an indigenous mixed broadleaf woodland with a Sitka spruce plantation of the same area in Scotland.

Table 1

Feature	Mixed indigenous woodland	Sitka spruce plantation
Tree species	12	1
Age structure	All ages, dead wood present	All planted in the same year
Planting density	Natural, variable	Close, 2 m spacing
Timber yield / m^3 ha^{-1} yr^{-1}	4	16
Bird species	38	9
Ground flora species	60	4

0 2 . 1 Use Table 1 to explain why the plantation has a much higher timber yield but far lower biodiversity.

[6 marks]

0 2 . 2 Explain why commercial plantations are felled before the trees reach full size.

[2 marks]

.....

.....

.....

.....

0 2 . 3 Explain why there is little demand today for coppiced or pollarded wood.

[2 marks]

.....

.....

.....

.....

10

0 3 . 1 Describe the ecological features of temperate broadleaf woodland and explain how woodland floor plants such as bluebells are adapted to them.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 3 . 2 Explain why an ancient woodland (existing before 1600) usually has higher biodiversity than a recently planted woodland.

[3 marks]

.....

.....

.....

Do not write outside the box

.....

.....

.....

.....

0 3 . 3 Describe **three** threats to UK broadleaf woodland.

[3 marks]

1 .....

.....

2 .....

.....

3 .....

.....

10

0 4 . 1 Explain the meaning of each of the following terms and explain why such species are given conservation priority.

[8 marks]

Do not write outside the box

EDGE species .....

.....

.....

.....

Endemic species .....

.....

.....

.....

Keystone species .....

.....

.....

.....

Flagship species .....

.....

.....

.....

0 4 . 2 Explain why the degree of population dispersal is considered when setting priorities.

[2 marks]

.....

.....

.....

.....

10

Do not write outside the box

0 5 . 1 Describe **two** case studies of biological control, one successful and one that caused problems.

[4 marks]

0 5 . 2 Explain how pheromone traps can be used in pest control.

[2 marks]

0 5 . 3 Explain why the use of antibiotics as growth promoters in livestock is a concern for human health.

[3 marks]

0 5 . 4 Give **one** way farmers can support crop pollinators.

[1 mark]

Do not write outside the box

10

0 6 . 1 Explain how each of the following environmental impacts of salmon farming arises and how it can be reduced.

[8 marks]

Overfishing of wild fish for feed .....

.....

.....

.....

Sea lice and disease .....

.....

.....

.....

Escapes affecting wild gene pools .....

.....

.....

.....

Organic waste pollution .....

.....

.....

.....

0 6 . 2 Explain why culling seals and cormorants near fish farms is controversial and give an alternative.

[2 marks]

.....

.....

.....

.....

10

Do not write outside the box

0 7 The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where S_1 and S_2 are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

Table 1

Term	Biomass / thousand tonnes
S_1	600
A	90
G	150
C	220
M	80

0 7 . 1 Calculate S_2 and state whether the stock is being fished sustainably.

[2 marks]

Answer ☐ thousand tonnes

0 7 . 2 Calculate the maximum catch that would have kept the stock constant.

[1 mark]

Answer ☐ thousand tonnes

0 7 . 3 Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult.

[3 marks]

.....

.....

.....

.....

.....

Do not write outside the box

Do not write outside the box

.....

.....

0 7 . 4 Describe how each of the following measures obtained from commercial catches can indicate overfishing.

[4 marks]

Catch per unit fishing effort .....

.....

.....

.....

Mean fish size and mean age .....

.....

.....

.....

10

0 8 A new pond was dug in a nature reserve. Table 1 shows the vegetation recorded at intervals over 60 years.

Table 1

Years after creation	Dominant vegetation	Water depth at centre / m
1	Single-celled algae, floating duckweed	2.0
5	Submerged pondweeds; reeds at margins	1.8
15	Reeds and reedmace over most of pond; water lilies	1.0
30	Marsh plants, sedges; willow and alder at edges	0.2
60	Willow / alder carr; oak seedlings	0 (soil waterlogged)

0 8 . 1 Name the type of sere shown in Table 1 and identify the pioneer species.

[2 marks]

.....

.....

.....

.....

0 8 . 2 Explain the changes shown in Table 1.

[5 marks]

0 8 . 3 Explain why a lithosere and a hydrosere in the same region end with the same climax community.

[2 marks]

0 8 . 4 Describe the early stages of a lithosere.

[1 mark]

10

Do not write outside the box

0 9 . 1 Give **two** natural processes that are driven by solar energy.

[2 marks]

1 .....

.....

2 .....

.....

0 9 . 2 Explain why natural processes can take place at low temperatures.

[2 marks]

.....

.....

.....

.....

0 9 . 3 Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

.....

.....

5

1 0 . 1 Evaluate the relative importance of ex-situ and in-situ methods in the conservation of endangered species.

[9 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Do not write outside the box

Do not write outside the box

Extra space

.....

.....

.....

.....

1 0 . 2 State what is meant by **rewilding**.

[1 mark]

.....

.....

10

Do not write outside the box

Write an essay on one of the following topics.

11 . 1 Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

OR

11 . 2 Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

END OF QUESTIONS